# 1. Clinical Question & Core Hypothesis

## Clinical Question

In MIBC, intratumoral immune structures (TLS/TILs) should enhance immunotherapy benefit, but many immune■infiltrated tumors fail to respond; what tumor-intrinsic features modulate whether local immune organization translates into checkpoint■blockade sensitivity? 2) Tier 1 basis: I built on CORE PATH 1 (TMB → immunogenicity) and CORE PATH 3 (MSH6 → MMR, mutational processes). TMB provides the neoantigen load axis while MSH6 status modulates the quality/distribution of those mutations and hence neoantigen clonality and immune visibility. 3) Tier 2 support: Noninvasive readouts (urine DNA methylation, CTCs) and trial/BCG biomarker biology suggest practical ways to detect the immune–mutation interaction over time; AI-based integration can harmonize heterogeneous signals for patient stratification. 4) Endpoint choice: I selected therapy_response (immune checkpoint blockade response) because the Ontologist's core question centers on differential immunotherapy outcomes linked to TMB/MMR biology and because therapy response is the mechanistic readout for neoantigen■driven immunity. 5) Deliberate exclusions: I excluded specific epigenetic mechanisms and broader metabolic modifiers (e.g., HDL-C, cross■cancer PRAD signals) from the core claim to keep the hypothesis focused on the mutation–immune interaction; urine methylation/CTCs/TP53/AI are placed as secondary modifiers for downstream detection or refinement rather than as core mechanistic drivers?

## Working Hypothesis

Primary: In muscle-invasive bladder cancer, TLS-like lymphoid aggregates predict immunotherapy benefit only when the tumor harbors a high clonal neoantigen burden and intact MSH6 function. Loss of MSH6 shifts the neoantigen landscape toward subclonal or lower quality neoantigens and may impede efficient antigen presentation, thereby blunting TLS-driven T cell priming and reducing response to checkpoint blockade despite high TMB. Consequently, TLS-high, TMB-high tumors with MSH6 deficiency will have inferior objective response rates and shorter progression-free survival compared with TLS-high, TMB-high, MSH6-intact tumors. Secondary, testable assertions: (a) among TLS-high tumors, the strongest predictor of response is the interaction of TMB and MSH6 status; (b) subclonal neoantigen burden and reduced MHC I pathway activity will mediate diminished response in MSH6-deficient tumors. These hypotheses will be tested using integrated genomic (WES for TMB, MSH6 sequencing/IHC), neoantigen clonality analyses, and antigen presentation readouts in ICI-treated MIBC cohorts, with noninvasive modifiers (urine DNA methylation, CTC signatures) explored as adjunct biomarkers.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | TMB, MSH6 |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**TMB:** TMB provides the neoantigen load axis while MSH6 status modulates the quality/distribution of those mutations and hence neoantigen clonality and immune visibility.

**MSH6:** MSH6 status modulates the quality/distribution of those mutations and hence neoantigen clonality and immune visibility.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|

| | | |
|---|---|---|
| Secondary | selected urine-DNA methylation biomarkers | Supporting modifier from knowledge graph |
| Secondary | circulating tumor cells (CTCs) | Supporting modifier from knowledge graph |
| Secondary | TP53 | Supporting modifier from knowledge graph |
| Secondary | AI-based biomarker translation | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** , and therapy response. Experimental plan: assemble a retrospective ICI-treated MIBC cohort with available tumor WES, MSH6 status, TLS assessment, and outcomes; compute TMB and clonal vs subclonal neoantigen burden; evaluate MHC I pathway activity (B2M/TAP

**Prediction 2:** and HLA LOH. Fit multivariable models including TLS, TMB, MSH6 and their interactions to predict ORR and PFS.

**Prediction 3:** In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions

- selected urine-DNA methylation biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- circulating tumor cells (CTCs): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.